Solving Recurrence Relations: Tree Method and Master Theorem

Part 1: Recursion Tree Method

Problem 1: $T(n) = 2T(n/2) + n$

Each level has cost n, and depth is $\log_2 n$. Total cost:

$$T(n) = n + n + n + \cdots + n = n \cdot \log_2 n = \Theta(n \log n)$$

Problem 2: $T(n) = 3T(n/2) + n$

Tree analysis:

Level 0: n Level 1: $3 \cdot (n/2) = 3n/2$ Level 2: $9 \cdot (n/4) = 9n/4$ Level i: $3^i \cdot (n/2^i)$

Total cost:

$$T(n) = \sum_{i=0}^{\log_2 n} 3^i \cdot \frac{n}{2^i} = n \cdot \sum_{i=0}^{\log_2 n} \left(\frac{3}{2}\right)^i$$

Geometric series with ratio > 1:

$$T(n) = \Theta(n \cdot (3/2)^{\log_2 n}) = \Theta(n^{\log_2 3}) \approx \Theta(n^{1.585})$$

Problem 3: $T(n) = 4T(n/2) + n^2$

Level i has: 4^i nodes each doing $(n/2^i)^2$ work:

$$\text{Total at level } i : 4^i \cdot (n^2/4^i) = n^2$$

All levels cost n^2, and there are $\log_2 n$ levels:

$$T(n) = n^2 \cdot \log_2 n = \Theta(n^2 \log n)$$

Problem 4: $T(n) = 2T(n/2) + n^2$

Level i: 2^i calls of $(n/2^i)^2 = n^2/4^i$ Total at level i: $2^i \cdot n^2/4^i = n^2/2^i$

Sum:

$$T(n) = \sum_{i=0}^{\log_2 n} \frac{n^2}{2^i} \leq 2n^2 = \Theta(n^2)$$

Problem 5: $T(n) = T(n/2) + T(n/4) + n$

This irregular recurrence can be visualized as: Each call splits into one size-$n/2$ and one size-$n/4$.

Let's estimate number of nodes:

- Tree has $O(\log n)$ height. - Each level cost is $\leq cn$. - Total work: $\leq cn \cdot \log n = \Theta(n \log n)$

Answer: $\boxed{\Theta(n \log n)}$

Part 2: Master Theorem

Master Theorem form:

$$T(n) = aT(n/b) + f(n)$$

Problem 1: $T(n) = 2T(n/2) + n$

Here: $a = 2,\ b = 2,\ f(n) = n$

$$n^{\log_b a} = n^{\log_2 2} = n \Rightarrow f(n) = \Theta(n) \Rightarrow \text{Case 2}$$

$$\boxed{T(n) = \Theta(n \log n)}$$

Problem 2: $T(n) = 4T(n/2) + n^2$

$a = 4,\ b = 2,\ f(n) = n^2$

$$n^{\log_2 4} = n^2, \quad f(n) = \Theta(n^2) \Rightarrow \text{Case 2} \Rightarrow \boxed{T(n) = \Theta(n^2 \log n)}$$

Problem 3: $T(n) = 2T(n/2) + n^2$

$a = 2,\ b = 2,\ f(n) = n^2$

$$n^{\log_2 2} = n, \quad f(n) = \Omega(n^{1+\varepsilon}), \varepsilon = 1$$

Check regularity:

$$2 \cdot f(n/2) = 2 \cdot (n/2)^2 = n^2/2 < cn^2,\ \text{for } c = 0.9 \Rightarrow \text{Case 3} \Rightarrow \boxed{T(n) = \Theta(n^2)}$$

Problem 4: $T(n) = 3T(n/4) + n \log n$

$a = 3,\ b = 4,\ f(n) = n \log n$

$$n^{\log_4 3} \approx n^{0.792},\ f(n) = \Omega(n^{0.792+\varepsilon})$$

Check regularity:

$$3f(n/4) = 3 \cdot \frac{n}{4} \cdot \log(n/4) = O(n \log n) \Rightarrow \text{Case 3} \Rightarrow \boxed{T(n) = \Theta(n \log n)}$$

Problem 5: $T(n) = 9T(n/3) + n$

$a = 9,\ b = 3,\ f(n) = n$

$$n^{\log_3 9} = n^2, \quad f(n) = O(n^{2-1}) = O(n) \Rightarrow \text{Case 1} \Rightarrow \boxed{T(n) = \Theta(n^2)}$$

Summary Table

Problem	Method	Final Answer
Tree 1	Tree	$\Theta(n \log n)$
Tree 2	Tree	$\Theta(n^{1.585})$
Tree 3	Tree	$\Theta(n^2 \log n)$
Tree 4	Tree	$\Theta(n^2)$
Tree 5	Tree	$\Theta(n \log n)$
Master 1	Master	$\Theta(n \log n)$
Master 2	Master	$\Theta(n^2 \log n)$
Master 3	Master	$\Theta(n^2)$
Master 4	Master	$\Theta(n \log n)$
Master 5	Master	$\Theta(n^2)$